

The Sound of Waves

We hear sounds all around: a dog's bark, a shoe dropping, rain on the roof. Sound is a form of energy that **vibrates**, or shakes, its surroundings. These vibrations travel in waves through the air like ripples in a pond.

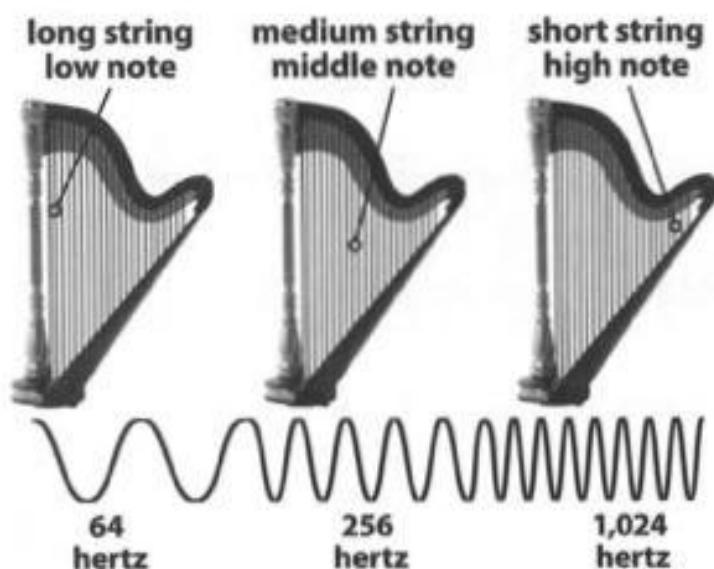
We primarily hear sound, but we can also see its vibrations. Stretch a rubber band between a finger and your thumb on one hand. Pluck the rubber band and watch it vibrate for a second as it makes a soft buzz. Change the **tension**, or how tightly the rubber band is pulled, and the buzz will change, too. That change in buzz is the rubber band's **pitch**. As you stretch the band more, the pitch goes up. As you stretch it less, it goes down.

The pitch depends on the speed of the sound waves. The waves travel back and forth very quickly, often hundreds of times per second. They move from one end of a stretchy, hollow, or solid material to the other. The greater number, or **frequency**, of trips per second, the higher the pitch. The less mass there is vibrating, the more trips sound waves can make. A string's mass is related to its length and width and to the material it is made of.

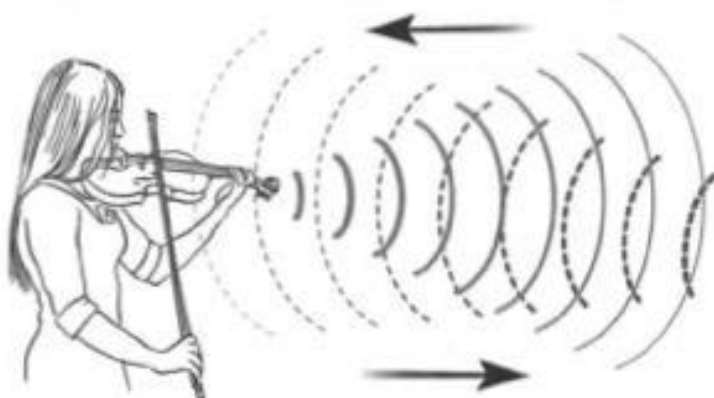
Sound travels through air, liquids, and solids. When sound waves hit surfaces, it affects the sound. Soft materials, such as drapes, **absorb** sound and make it quieter. Hard, smooth materials, such as tile, **reflect** sound and make it louder. When sound waves enter a space with sides, they **reverberate**, or bounce around, increasing the length of time you hear it.

Stringed instruments have strings that are plucked, strummed, or played with a bow to make the different pitches. Some instruments, such as a harp or a koto, have one string for each note. Stringed instruments with a neck, such as a banjo or a sitar, use a limited number of strings, which the player can shorten by pressing at different places along the neck to change the pitch.

STEM Challenge: Stringed Instrument

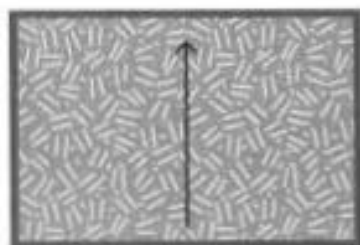


Three different lengths of string make waves of different **frequencies**. Frequencies are measured in **hertz**. One hertz is one round-trip of a sound wave per second. The longest string's wave frequency is slow; it is a low note. The shortest string's waves move fast; its note is quite high. The middle string makes medium-sized waves and a medium-sounding pitch.



Round-trip of waves along a vibrating violin string

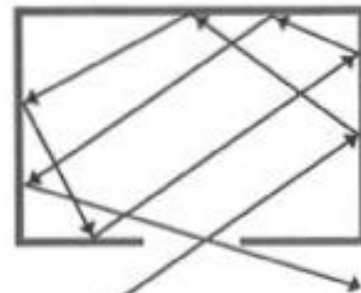
The sound waves zoom from one end of the string to the other, back and forth many times per second. The ends are the two places where the string is attached or pressed down. You can see the string vibrate as the sound waves push invisible air molecules.



Soft materials
absorb sound.



Hard materials
reflect sound.



Sound **reverberates**
between hard surfaces.

Stringed Instrument

Stringed Instrument

Challenge: Design and create a stringed instrument with at least 4 strings that can play a variety of pitches.

Testable goal: The instrument produces at least 8 different pitches.

Research: Look at pictures of stringed instruments, such as guitars, violins, sitars, harps, kotos, and hammered dulcimers. Notice how they are constructed. Think about how science, technology, engineering, and math are used to create a stringed instrument.

Brainstorm: Draw one or more design ideas for a stringed instrument. There are many different ways to complete this challenge. Be creative!

A large, empty rectangular box with a rounded border, intended for students to draw their design ideas for a stringed instrument.

STEM Challenge:

Stringed Instrument

Suggested Materials List

Items for each group

☐ scissors

☐ pencil

☐ ruler

Items for the whole class

☐ paper towel tubes

☐ toothpicks

☐ gift wrap tubes

☐ stapler

☐ craft sticks

☐ clay

☐ empty tissue boxes

☐ masking tape

☐ empty shoe boxes

☐ glue

☐ rubber bands

☐ _____

☐ thin wire

☐ _____

☐ fishing line

☐ _____

☐ dental floss (unwaxed)

☐ _____

☐ thin elastic

☐ _____

☐ rubber erasers

☐ _____

Items for testing

☐ various mallets: spoons, pencil erasers, etc.

1 Plan

Think about the materials available to you. How can you use the materials to create a stringed instrument that produces at least 8 pitches?



2 Create

Use the materials you have to create a stringed instrument. Follow your design as closely as possible.



3 Test

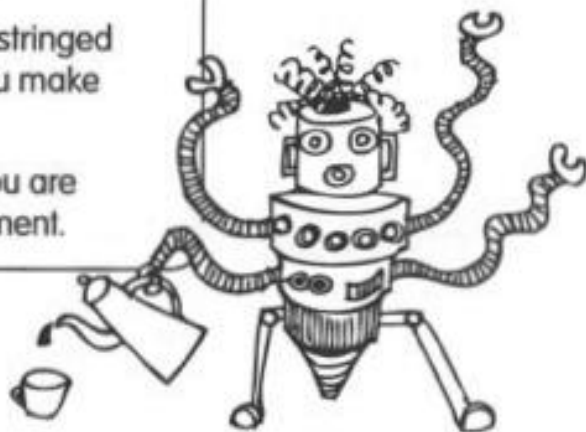
Play your stringed instrument. Does it produce at least 8 different pitches? Will it play even more pitches? Do you like the tone? How far away can the pitches be heard?



4 Evaluate and Revise

Evaluate the performance of your stringed instrument. What revisions can you make to improve its performance?

Repeat the design process until you are satisfied with your stringed instrument.



1 Plan: Describe and draw your design. List your materials.



2 Create



3 Test: Describe and draw your results.



4 Evaluate

Stringed Instrument

1 Redesign: Describe what you will change. Draw your new design.



2 Revise or re-create



3 Test: Describe and draw your results.



4 Evaluate